Solar Energy Forecasting Using Machine Learning

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Abstract

Accurate solar power forecasting is essential for efficient energy management and grid stability. This project presents a machine learning-based approach to predict solar power generation using environmental and time-based features. A Random Forest Regression model was trained on solar plant data, achieving strong predictive performance with a high R^2 score and low error metrics. Experimental results confirm that irradiation is the dominant predictor, while engineered time-series features improve short-term forecasting.

1 Introduction

With the rapid adoption of renewable energy, predicting solar power output has become crucial for optimizing grid operations and reducing uncertainty in energy supply. Machine learning models enable accurate forecasting by learning patterns from historical environmental data.

This project builds a predictive system to estimate solar energy output using weather and temporal features.

2 Problem Statement

The objective is to develop a machine learning model capable of predicting solar power generation based on irradiation, temperature, and time-based features to support better energy planning.

3 Dataset Description

The dataset contains solar plant operational data including:

- Solar power generation

- Irradiation levels
- Ambient temperature
- Module temperature
- Timestamp information

Additional features such as hour, month, lag variables, and rolling means were engineered.

4 Methodology

4.1 Data Preprocessing

- Timestamp converted to datetime
- Datasets merged
- Missing values handled
- Feature engineering (lag and rolling statistics)

4.2 Model Selection

Random Forest Regression was selected because of:

- Ability to model non-linear relationships
- Robustness to noise
- High performance on structured data

5 Model Evaluation

5.1 Evaluation Metrics Output

```
model = RandomForestRegressor(
    n_estimators=200,
    max_depth=12,
    random_state=42,
    n_jobs=-1
)

model.fit(X_train, y_train)

preds = model.predict(X_test)

mae = mean_absolute_error(y_test, preds)
rmse = np.sqrt(mean_squared_error(y_test, preds))
r2 = r2_score(y_test, preds)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2:", r2)
```

```
MAE: 10573.810102716383
RMSE: 21207.710057670523
R2: 0.9323337701435448
```

Figure 1: Model Performance Metrics Output

The model achieved:

- MAE = 10573.81
- RMSE = 21207.71
- $R^2 = 0.93$

5.2 Metric Definitions

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Measures average prediction error magnitude.

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Penalizes larger errors more heavily.

R^2 Score

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

An R^2 of 0.93 indicates the model explains about 93% of variance in solar power output.

6 Results and Visualization

6.1 15-Minute Ahead Forecast

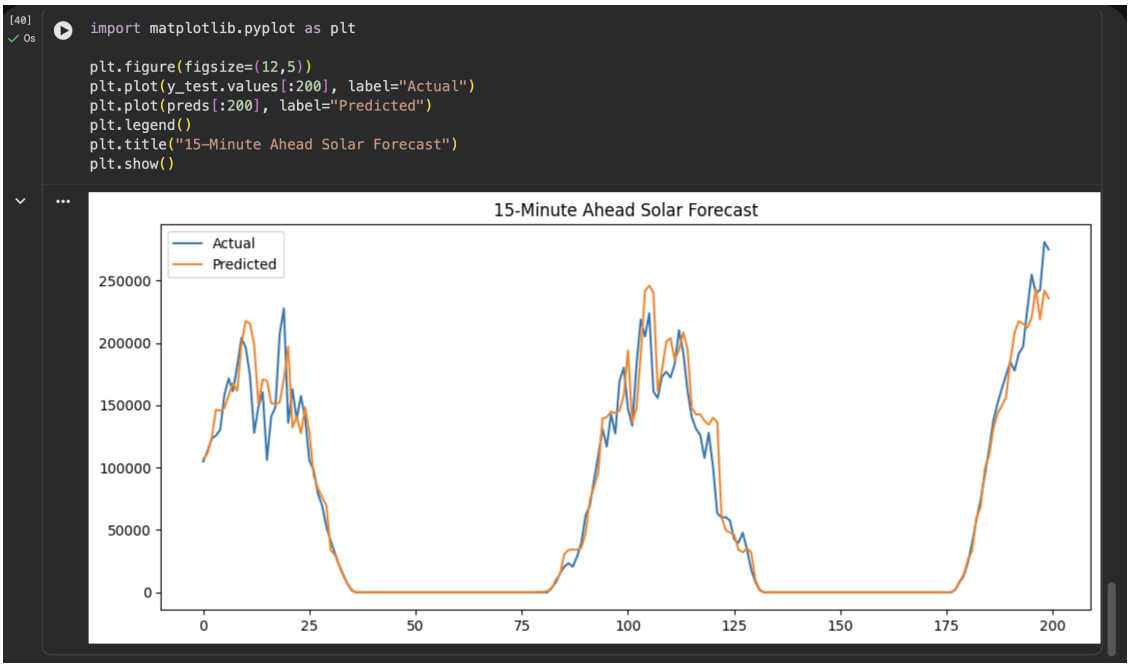

Figure 2: Actual vs Predicted Solar Output

The prediction plot shows that the model accurately captures daily generation patterns and peak production hours.

6.2 Feature Importance

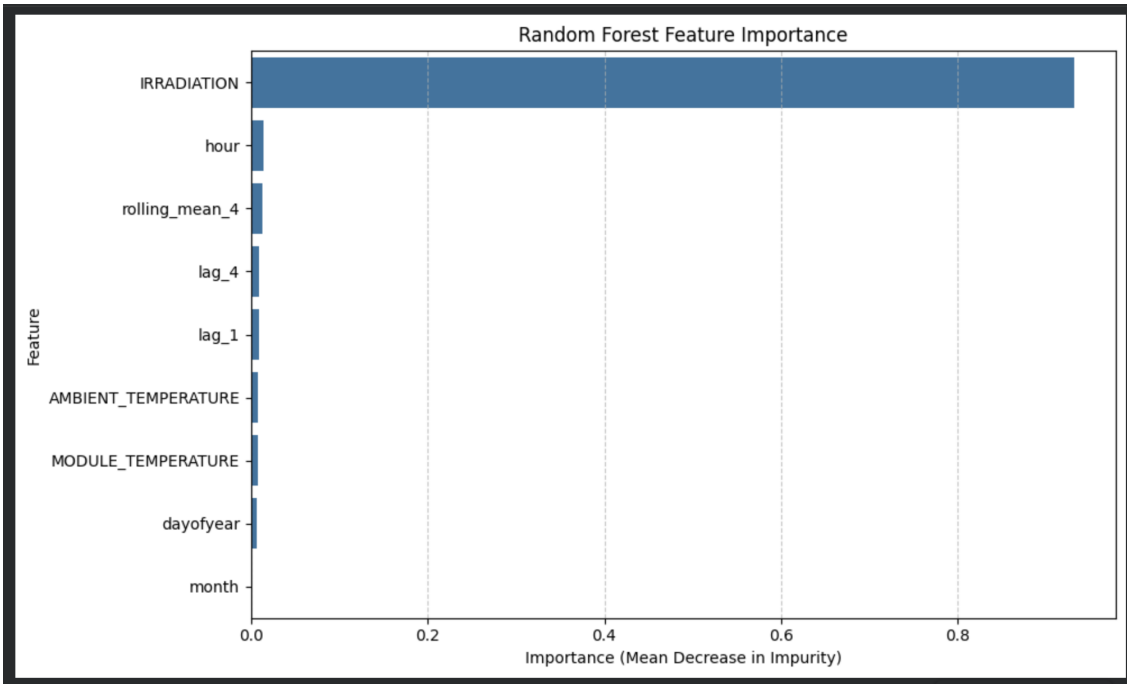

Figure 3: Random Forest Feature Importance

Irradiation dominates feature importance, confirming domain knowledge that solar intensity drives power output.

6.3 Discussion

Key observations:

- Strong alignment between predicted and actual values
- Accurate peak prediction
- Stable performance across different time periods

7 Model Optimization

Performance improvements were achieved through:

- Feature engineering
- Hyperparameter tuning
- Removal of low-impact features

8 Deployment

The trained model was deployed via a Streamlit application allowing real-time solar generation prediction.

9 Conclusion

This project demonstrates the effectiveness of machine learning in solar energy forecasting. The Random Forest model achieved strong accuracy and can support operational decision-making in renewable energy systems.

10 Future Work

- Real-time weather API integration
- Cloud deployment
- Advanced models such as LSTM and XGBoost

11 Team Contribution

- Ved Pawar — Project Lead
- Aviral Mishra — Data and Quality Lead
- Naitik Pandey — Report and Evaluation Lead
- Samarth Khera — Analysis Lead